LLJ Cards: Evaluating Creativity

Subsection	Q#	Answer
CONTEXT		
Application	1	The project is centered aournd the evaluation of Large Language models for creativity. Specifically, we tests LLMs capabilities for divergent thinking in open problem with goal-oriented tasks and we use the LLM judge to measure aspects of Originality, Surprise, and Relevance in the produced texts.
Application	2	The objective of divergent thinking in product design is to generate as many ideas as possible for a specific need. As such ideas can be potentially non-ethical when produced. If the LLM-judge rates an idea in such a way that it is considered highly relevant, original, and surprising, this could create a risk of bias leading to the adoption of potentially dangerous ideas that might have been rejected had it not been for that judge's evaluation.
Task	3	The tasks consist in generating as many ideas as possible for a given brief (Generate as many ideas as possible for "a strategy to mitigate climate change"). Then the judge is used to evaluate these ideas.
Task	4	The task is not evaluating LLM creativity. Creativity is a process which combines divergent thinking, idea elaboration and development, convergent thinking and is perceived differently for various domain (creativity in arts is not the same as creativity for scientific discovery). Therefore, the tasks do not evaluate how LLMs creative capabilities influence the full design of a product.
Task	5	If products are designed with AI, every person using these products might be impacted using these specific products. With the problems of homogenization in AI outcomes, this could lead to severe cultural or ethical impact on populations for example.
Cases	6	Generated design ideas for divergent-thinking tasks.
Cases	6.1	Author-created dataset for divergent thinking on goal-oriented product design.
Cases	6.2	Creativity
Cases	6.3	It was created by (anonymized) to evaluate LLMs capability to generate as many ideas as possible for products (bookmarks), services (lemonade stand), strategies (climate change), and process (recruitment).
Cases	6.4	It is limited to these tasks and to divergent thinking only. The dataset also includes ethically ambiguous tasks used to evaluate the values of LLMs generated ideas.
Cases	6.5	North American Values
Cases	6.6	This evaluation is independent of the evaluation purpose of the LLM-judge, and it should not be re-used for such purpose. The judge should be re-used only for evaluating idea generation and restricted dimension of originality, surprise, and relevance.
Stakeholders	7	The LLJ is used in an academic setting. The model used is the "Qwen/Qwen3-30B-A3B-Instruct-2507" open-source model from Hugging Face. The dataset was created in 2025 and the LLJ-judge study was performed in 2026. The project is funded by the authors.
Stakeholders	8	
Stakeholders	9	

Subsection	Q#	Answer
EVALUATION CRITERIA		
Systematized Concept	10	We are trying to evaluate the "creativity" of a ensemble of design solution for a same problem. The ideas are LLM produced through prompting strategies such as "generate as many diverse ideas as possible for: A bookmark".
Systematized Concept	11	If the ensemble of ideas is diverse, and that each individual idea is original, surprising, and relevant to the tasks then the overall creativity if the system output should be considered as high. If all dimensions are low, the system is evaluated as less creative. Overall, we do not assess creativity if one metric (Originality) is high and another (Surprise) is low.
Systematized Concept	12	We believe that with the essential problems of homogenization and as LLMs and Generative AI is more and more involved in creative activity, this concept should be better evaluated to understand how it influences arts, narration, scientific discovery, product design and other creative activities.
Operationalization	13	We operationalize the evaluation of creativity using 3 distinct criteria: Originality, Surprise, and Relevance of the proposed individual ideas. We also add the diversity of the ensemble of ideas generated for a same product.
Operationalization	14	We chose this operationalization to measure creativity as it is a well supported decomposition of this concept in management, creativity and entrepreneurship literature. The main references for this decomposition are the two following articles: Boden, M. A. (2004). The creative mind: Myths and mechanisms. Routledge. / Amabile, T. M. (1983). The social psychology of creativity: A componential conceptualization. Journal of personality and social psychology, 45(2), 357.
Operationalization	15	These concepts originated in the 1980s within the social sciences and have since been validated through numerous studies. Their operationalization through classic divergence metrics is now well established: Silvia, P. J., Winterstein, B. P., Willse, J. T., Barona, C. M., Cram, J. T., Hess, K. I., ... & Richard, C. A. (2008). Assessing creativity with divergent thinking tasks: Exploring the reliability and validity of new subjective scoring methods. Psychology of Aesthetics, Creativity, and the Arts, 2(2), 68. Grace, K., Maher, M. L., Fisher, D., & Brady, K. (2015). Data-intensive evaluation of design creativity using novelty, value, and surprise. International Journal of Design Creativity and Innovation, 3(3-4), 125-147.
Operationalization	16	These criteria are not necessarily tasks dependent, but they have been applied to their most common used case: ideation session for problem solving.
Operationalization	16.1	
Limitations	17	As creativity is intimately related to one's knowledge, creativity for the whole of humanity (H-Creativity) differs from creativity for a specific society (S-Creativity) or an individual person (P-Creativity). These metrics therefore capture only a particular notion of creativity defined by the underlying knowledge representation. This limitation is especially important when evaluating LLM creativity, as the knowledge encoded in such models is itself biased toward certain cultures and societal values. We fully acknowledge the limitations of our validation process, which was constrained by the timing of the rebuttal period. We encourage the community to further strengthen the validity of these evaluations before adopting our LLJ model and to restrict its use to similar cases. We also emphasize that this model should not be used beyond the scope of this paper, namely measuring the correlation between LLJ outputs and established creativity metrics for academic research purposes only.
Limitations	18	North American Values

Subsection	Q#	Answer
LLJ PIPELINE		
Set-Up	19	Evaluation
Set-Up	20	LLJ
Set-Up	21	Free text generation of ideas. Here are two examples of ideas evaluated by the LLJ: "The Money Pit" Jet: A luxurious private jet in a flamboyant, iridescent green, adorned with gold accents and a perpetually open money chute on its underside. It represents unchecked wealth and environmental extravagance that causes hidden harm. "Expanded Bicycle Infrastructure & Secure Storage": Invest in more dedicated bike lanes, secure covered bike racks, and bike repair stations across campus. This makes cycling a more attractive and viable commuting option.
Set-Up	22	The evaluation uses a pointwise setup in which each idea is assessed independently. Every idea is rated on three dimensions, namely originality, surprise, and relevance. Scores are assigned using a Likert scale ranging from 1 (low) to 5 (high) for each criterion.
Model	23	Qwen3-30B-A3B-Instruct-2507 selected as a compromise between budget and performance. This version is available on Hugging Face: https://huggingface.co/Qwen/Qwen3-30B-A3B-Instruct-2507. Additional details regarding the model architecture, training process, capabilities, and usage recommendations are provided on the model card.
Model	23.1	Qwen
Model	23.2	Qwen3-30B-A3B-Instruct-2507
Model	23.3	30B
Model	23.4	July 2026
Model	23.5	tensor_parallel_size=2, gpu_memory_utilization=0.75, max_model_len=4096, dtype=auto, enforce_eager=True, temperature=0.7, max_tokens=512, top_p=1.0, batch_size=10, k=5
Model	23.6	Compromise of budget and performance.
Reproducibility	24	The model requires 2 GPU RTX8000 with at least 48GB of Memory per GPUs to run as well as an 8-core CPUs with 254GB of RAM.
Reproducibility	24.1	No
Reproducibility	24.2	Yes
Reproducibility	24.3	Yes
Budget	25	
Budget	25.1	
Budget	25.2	Model requires two RTX8000 GPUs with at least 48GB memory per GPU and one 8-core CPU with 254GB RAM.
Prompt	26	Expert evaluator of idea novelty and appropriateness.
Prompt	26.1	You are an expert evaluator of idea novelty and idea appropriatness.
Prompt	26.2	Assess the novelty of the following idea along four conceptually distinct dimensions of novelty defined below. Each dimension relies on a different frame of reference, so evaluate them independently and avoid conflating them. For each dimension, assign a score from 1 (very low) to 7 (very high). Assess the appropriateness of each idea relative to the brief context using a binary label: 1 (appropriate) or 0 (not appropriate).
Prompt	26.3	Newness, Uniqueness, Difference, Surprise, Appropriateness.

Subsection	Q#	Answer
Prompt	26.4	Definitions provided in the prompt. The definition of the criteria was extracted from the relevant literature by one of the authors from the paper, who is a specialized researcher in management of creativity and innovation. The definitions can be identified from the Amabile and Boden references provided earlier. Dimension 1 - Newness: The extent to which the idea introduces properties, terms, or knowledge that had never been observed or made public before, providing a genuine gain in information that expands the existing knowledge base. Anchor 1 (Low): Restates already-known, publicly available knowledge; no informational gain. Anchor 7 (High): Introduces genuinely new knowledge, terms, or properties never previously observed. Dimension 2 - Uniqueness: The extent to which the idea is uncommon, rare, or singular relative to a pool of ideas or existing categories, standing out from a prototypical exemplar. Anchor 1 (Low): Highly frequent; closely matches the prototype or typical response. Anchor 7 (High): Rare and singular; clearly distant from the prototypical category. Dimension 3 - Difference: The degree to which the idea differs from and diverges from prior or existing designs, objects, or entities. Anchor 1 (Low): Closely resembles existing designs or objects; minimal divergence. Anchor 7 (High): Strongly atypical; diverges markedly from the core aspects of existing entities. Dimension 4 - Surprise: The extent to which the idea violates expectations about an object's likely attributes, combinations, or recombinations. Anchor 1 (Low): Fully expected; obvious combination of attributes; conforms to domain trends. Anchor 7 (High): Highly unexpected; subverts implied rules; counterintuitive; defies expectations. Appropriateness: An idea remains faithful to the original intent, topic, and constraints of the brief. Label 1 (Appropriate): Adequately addresses directly or indirectly the specific topic and keywords in the brief. Label 0 (Not Appropriate): Fails to adequately address the specific topic and keywords in the brief. Appropriateness is independent of originality; an idea may be unconventional yet still appropriate if it remains faithful to the brief's intent.
Prompt	26.5	
Prompt	26.6	Bavatto (2022), Fiorineschi & Rotini (2021), Harvey & Berry (2023), Deichmann et al. (2024), Carichon et al. (2025).
Prompt	26.7	Brief/context and generated idea.
Prompt	26.8	Scores (1-7) for four novelty dimensions and an appropriateness label (0/1).
Score	27	The score is outputted on a Likert scale for each criterion from 1 to 5 (1 being poorly scored on the given criteria and 5 high on the criteria). The judge is running 5 times and the average of these 5 runs is kept computing the final score.
Score	28	Most studies evaluating creativity with human subject employs the same pattern: Evaluating the idea on the same criteria on a Likert scale scheme and averaging human response for the given criteria. Here is a reference for the protocol: Sarkar, P. Chakrabarti, and J. Gero. "Studying engineering design creativity". Workshop on Studying Design Creativity. 2008.

Subsection	Q#	Answer
LLJ EVALUATION		
Evaluation	29	The authors evaluated manually on the same dataset on a sample of 30 ideas the quality of the LLJ outputs to confirm the metrics were evaluating the proper concept. We also compare the evaluation to metrics based on semantic distance, reported in our paper (anonymized) to compare the convergent similarity of the judge. We are not evaluating the model beyond this.
Evaluation	29.1	Semantic distance metrics for diversity, novelty, rarity, and relevance.
Evaluation	29.2	Manual expert assessment of 30 sampled ideas.
Evaluation	30	We are not presenting any other results, only aggregated metrics
Evaluation	31	The authors, which are expert on creativity and management science, manually assessed the validity of the scores.
Evaluation	32	The evaluation process is limited as we cannot compare the LLJ outputs to any human evaluations. As we mention to compare the LLJ results to human we would need to control that both the human judge and the LLMs possess the same knowledge space for each design evaluated which is not doable for human evaluation as of now. Therefore, except for controlling the match between the definitions of operationalization and the output, the evaluation of LLJ in our application case is limited. One improvement would be to report LLJ evaluation on other classic divergent thinking tasks as the DAT or the TTCT.
Evaluation	33	The evaluation is static and represents a one-shot evaluation to demonstrate the correlation between semantic based distance and the results extracted from a LLJ given very specific and well-established definitions of creativity.
Evaluation	33.1	
Ethical Considerations	34	We could promote non original and non diverse outputs with wrong evaluation contributing further ahead to the wrong evaluation of LLM homogenization phenomenon.
Ethical Considerations	35	The LLJ results could amplify existing bias compared to semantic and superficial metrics such as BLEU because the LLJ could over-estimate the originality, surprise, and relevance of its results generated by LLMs compared to solutions produced by humans.
Ethical Considerations	36	As I said, the lack of reliability of this metrics compared to human based judgments that are explicitly related to people cultural background could lead to potential cultural homogenization and lack of creativity in solution design.

Subsection	Q#	Answer
LLJ PROTOTYPING		
Study Set-Up	37	
Study Set-Up	37.1	The authors semantic based distance metrics to measure diversity, novelty, rarity and relevance of idea as defined by previous works (Mohseni, Maryam, and Mary Lou Maher. "A framework for exploring computational models of novelty in unstructured text." Proceedings of the 6th International Conference on Information System and Data Mining. 2022.). However, the authors decided to reinforce their analysis by using LLJ evaluation and measuring the correlation between these metrics and LLJ evaluation of the same criteria. The prototyping phase including: 1. selecting the criteria from the relevant literature. 2. Defining the concepts operationalizing the measure of the criteria. 3. Manually reviewing the appropriateness of these results with expert evaluators from the literature on creativity and innovation. 4. Selecting models for LLJ evaluation with testing multiple models.
Study Set-Up	37.2	Criteria selection, concept operationalization, expert annotation, semantic distance metrics, and testing multiple LLJ models.
Study Set-Up	37.3	
Study Set-Up	37.4	
Data	38	We manually evaluated a sample of 50 ideas randomly extracted from (anonymized) dataset to measure the appropriateness of the results.
Data	39	No
Design	40	We tested LLama3.1 8B, Qwen 2.5 7B, Qwen 2.5 32B, GPT 5.5 Mini, Qwen3 8B and Qwen3 30B for the LLJ evaluation. We retained Qwen3-30B to compromise between budget constraints and performances. GPT 5.5 Mini was the best performing model.
Design	41	No, we do not report manual evaluation for this phase of the process.
Validation	42	No, we did not test the model for validity beyond the extent of the 50 samples evaluated by the expert authors.
Validation	43	No, we did not test the model for validity beyond the extent of the 50 samples evaluated by the expert authors.
Refinement	44	Yes, several changes were performed to modify the prompts.
Refinement	45	We stopped modification when the expert authors evaluated that the results from the LLJ from the various models were respecting properly the provided definitions on the 50 ideas sample.